

ECE 4375, Final Exam Review Material

1. Consider the differential equations,

$$\ddot{y}(t) - \dot{y}(t) + 6y(t) = u(t)$$

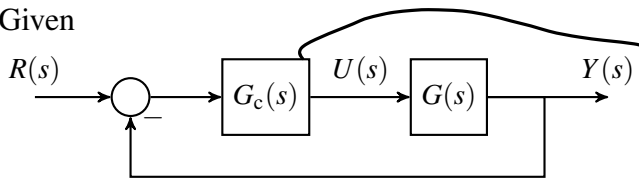
$$\ddot{y}(t) + 5\dot{y}(t) + 2y(t) - \dot{u}(t) + u(t) = 0$$

Find the transfer function, $\frac{Y(s)}{U(s)}$. What assumptions must be made regarding the initial conditions? Is the system stable? What are the initial and final values of the system when the input is a unit step? What is the system type of $\frac{Y(s)}{U(s)}$?

Now, take the $\frac{Y(s)}{U(s)}$ transfer function and apply unity negative feedback around it to produce $\frac{Y(s)}{R(s)}$. How has the feedback changed the stability? What are the new initial and final values when the input is a unit step? What is the system type of $\frac{Y(s)}{R(s)}$?

Finally, transform the $\frac{Y(s)}{R(s)}$ transfer function back into a differential equation.

2. Given



$$G_c = K_p + \frac{K_i}{s} + K_d s$$

$$\text{set } K_p, K_i, K_d = 1$$

When

A. $G(s) = \frac{s^2 + 2s + 2}{(s^2 - 11s + 24)(s^2 + 7s)}$

$$G_c = 1 + \frac{1}{s} + s$$

B. $G(s) = \frac{s - 9}{(s - 3)(s^2 - 6s + 5)(s^2 + 2s + 5)}$

$$G_c = s + 1 + s^2$$

C. $G(s) = \frac{1}{(s + 1)(s + 10)}$

$$G_c = s^2 + s + 1$$

D. $G(s) = \frac{s(s + 0.1)}{(s + 1)(s + 10)}$

$$\rightarrow E) \quad G(s) = \frac{s + 1}{s^2(s + 10)^2}$$

E. $G(s) = \frac{s + 1}{s^2(s + 10)^2}$

$$G_c G(s) = \frac{(s + 1)(s^2 + s + 1)}{s^2(s + 10)^2}$$

F. $G(s) = \frac{900}{s^2 + 15s + 900}$

Perform the following tasks for $G_c = K_p$

1. Determine the BIBO stability with the Routh Array
2. The initial and final values for (i) impulse input and (ii) step input
3. Sketch the Root Locus *When appropriate* find:
 - (a) σ_i
 - (b) Θ_k
 - (c) BA and BI
 - (d) K_{BA} and K_{BI}
 - (e) critical gain for stability, K_{cr}
4. Determine the frequency response parameters (gain margin, phase margin, bandwidth, and low frequency gain) when $G_c = 1$
5. Determine $e_p(\infty)$, $e_v(\infty)$, and $e_a(\infty)$

Repeat for $G_c = K_p + \frac{K_i}{s}$

Finally, set $G_c = K_p + \frac{K_i}{s} + K_d s$ and solve 2, 3, and 5.

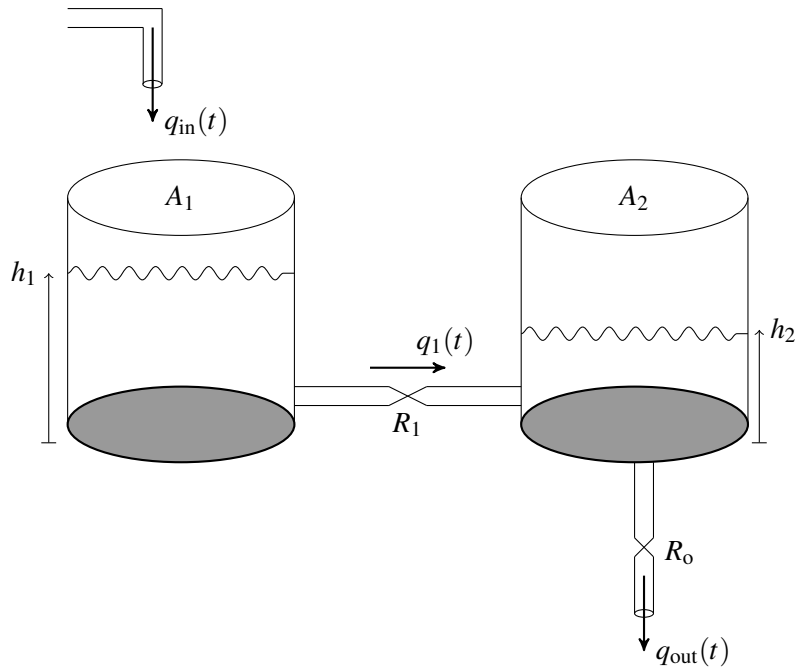
3. Given the 2nd order model for a DC-servo motor, $\frac{\omega_m(s)}{V_a(s)}$ where:

1. $R_a = 1.1$
2. $L_a = 0.0001$
3. $K_m = 12$
4. $B = 0.75$
5. $J_T = 0.85$
6. $K_b = 0.125$

Find the transfer functions $\frac{I_a(s)}{V_a(s)}$. What is the steady-state current given a step input?

4. Given the two tank system below, where:

A_1	=	3
A_2	=	6
R_1	=	1
R_o	=	2



Find:

- A. The block diagram of the system where the input is $Q_{in}(s)$ and the output is $Q_{out}(s)$
- B. The transfer function, $\frac{H_2(s)}{Q_{in}(s)}$

use base block diagrams (manipulation)
and solve

1. Consider the differential equations,

a. $\ddot{y}(t) - \dot{y}(t) + 6y(t) = u(t)$

b. $\ddot{y}(t) + 5\dot{y}(t) + 2y(t) - \dot{u}(t) + u(t) = 0$

Find the transfer function, $\frac{Y(s)}{U(s)}$. What assumptions must be made regarding the initial conditions? Is the system stable? What are the initial and final values of the system when the input is a unit step? What is the system type of $\frac{Y(s)}{U(s)}$?

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Finally, transform the $\frac{Y(s)}{R(s)}$ transfer function back into a differential equation.

(b) $\ddot{y}(t) + 5\dot{y}(t) + 2y(t) - \dot{u}(t) + u(t) = 0$

$\mathcal{L} \rightarrow s^2 Y(s) + 5sY(s) + 2Y(s) - sU(s) + u(s) = 0$

$s^2 Y(s) + 5sY(s) + 2Y(s) = sU(s) - u(s)$

$Y(s) [s^2 + 5s + 2] = U(s) [s - 1]$

$\frac{Y(s)}{U(s)} = \frac{s-1}{s^2 + 5s + 2}$

$$s^2 \left| \begin{array}{cc} 1 & 2 \\ s & 5 \\ s^0 & K_{31} \end{array} \right|$$

$$K_{31} = \frac{10 - 0}{5} = 2$$

No sign changes, so stable

(a) $\ddot{y}(t) - \dot{y}(t) + 6y(t) = u(t)$

$s^2 Y(s) - sY(s) + 6Y(s) = U(s)$

$Y(s) [s^2 - s + 6] = U(s)$

$\frac{Y(s)}{U(s)} = \frac{1}{s^2 - s + 6}$

Type 0

Routh Array

$$s^2 \left| \begin{array}{cc} 1 & 6 \\ s & -1 \\ s^0 & K_{31} \end{array} \right|$$

$K_{31} = \frac{-1(6)}{-1} = 6$

sign change, so unstable

Unit Step

$$u(t) = 1, \quad U(s) = \frac{1}{s}$$

$$\text{IVT: } y(0) = \lim_{s \rightarrow \infty} sY(s) = \lim_{s \rightarrow \infty} s \left[\frac{1}{s^2 - s + 6} \right] \cdot \frac{1}{s} \Big|_{s \rightarrow \infty}$$

$$= \lim_{s \rightarrow \infty} \left[\frac{1}{s^2 - s + 6} \right] \Big|_{s \rightarrow \infty} = \frac{\frac{1}{s^2}}{1 - \frac{1}{s} + \frac{6}{s^2}} \Big|_{s \rightarrow \infty} = \frac{0}{1} = \underline{0}$$

$$\text{FVT: } y(\infty) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} \left[\frac{1}{s^2 - s + 6} \right] \Big|_{s \rightarrow 0} = \underline{\frac{1}{6}}$$

$$\frac{Y(s)}{R(s)} = \frac{G}{1+GH} = \frac{\frac{1}{s^2 - s + 6}}{1 + \frac{1}{s^2 - s + 6}} = \frac{1}{s^2 - s + 7}$$

Type 0

Routh Array

$$\begin{array}{c|cc} s^2 & 1 & 7 \\ s & -1 & 0 \\ 7 & K_3 & 0 \end{array}$$

$$K_3 = \frac{-7}{-1} = 7$$

sign change; system is unstable

No change from previous

$$\text{IVT: } y(0) = \lim_{s \rightarrow \infty} sY(s) = \lim_{s \rightarrow \infty} \frac{1}{s^2 - s + 7} \Big|_{s \rightarrow \infty} = \frac{\frac{1}{s^2}}{1 - \frac{1}{s} + \frac{7}{s^2}} \Big|_{s \rightarrow \infty} = \frac{0}{1} = 0$$

$$\text{FVT: } y(\infty) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} \frac{1}{s^2 - s + 7} \Big|_{s \rightarrow 0} = \frac{1}{7}$$

$$\frac{Y(s)}{R(s)} = \frac{1}{s^2 - s + 7} \Rightarrow Y(s)[s^2 - s + 7] = R(s)$$

$$s^2 Y(s) - s Y(s) + 7 Y(s) = R(s)$$

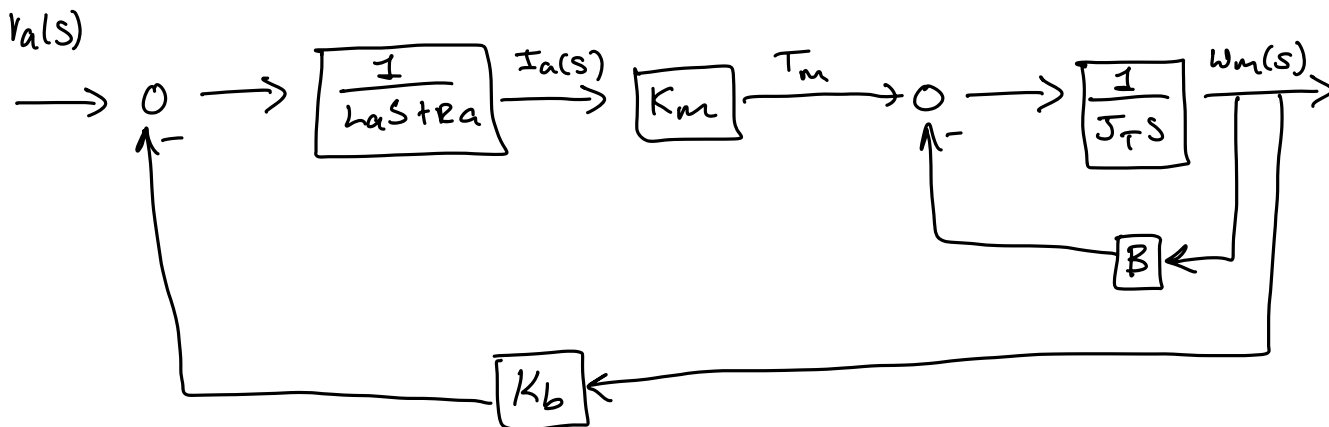
$$\ddot{y}(t) - \dot{y}(t) + 7y(t) - r(t) = 0$$

3. Given the 2nd order model for a DC-servo motor, $\frac{\omega_m(s)}{V_a(s)}$ where:

1. $R_a = 1.1$
2. $L_a = 0.0001$
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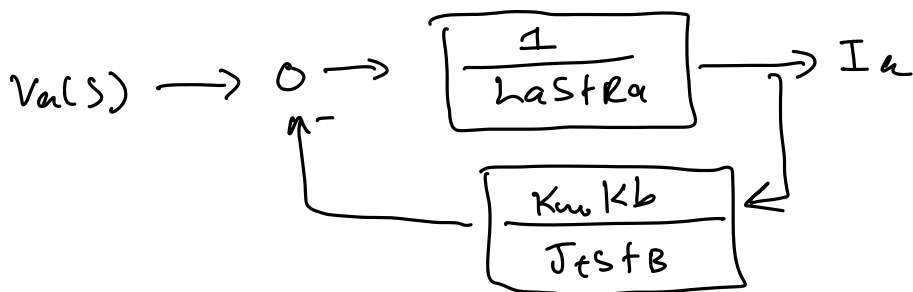
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DC MOTOR Block Diagram



$$\frac{\omega_m}{V_a} = \frac{K_m}{L_a J_T s^2 + (L_a B + R_a J_T) s + R_a B} \quad K_m K_b$$

want $I_a(s)$, so set $\omega_m(s) = 0$



$$\frac{I_a(s)}{V_a} = \frac{\frac{1}{L_a s + R_a}}{1 + \left(\frac{1}{L_a s + R_a} \right) \left(\frac{K_m K_b}{J_t s + B} \right)}$$

$$\frac{I_a(s)}{V_a} = \frac{J_t s + B}{L_a s^2 + s^2 + (R_a J_t + L_a B) s + R_a B + K_m K_b}$$

$$\frac{I_a}{V_a} = \frac{0.85 s + 0.75}{85 \times 10^{-6} s^2 + 0.935 s + 2.325}$$

$$I_{ss} = \lim_{s \rightarrow 0} s \left(\frac{0.85 s + 0.75}{85 \times 10^{-6} s^2 + 0.935 s + 2.325} \right) \left(\frac{1}{s} \right)$$

$$I_a(t \rightarrow \infty) = \frac{0.75}{2.325} = \underline{0.322 \text{ A}}$$